Altogether, our findings show that for various applications mode-locked femtosecond lasers may be replaceable by compact, cost-effective microchip lasers emitting single-mode sub-ns pulses in the UV/VIS wavelength range.

5.8. Consequences for laser ablation and breakdown spectroscopy in liquids

A deeper mechanistic understanding of optical breakdown in bulk liquid also helps to understand the rules governing plasma growth and hydrodynamic phenomena in laser ablation in liquids (LAL) for nanoparticle generation [71-81] and in underwater laser-induced breakdown spectroscopy (LIBS) for remote analysis of target constituents [82-89]. LAL is an ecofriendly and versatile approach that reduces the need for additional purification steps. It has been successfully employed to synthesize nanoparticles from metals, oxides semiconductors and organic materials directly in water and organic liquid media, making it attractive for applications requiring high-purity colloidal nanoparticles [72,316]. Underwater LIBS is of major importance for deep-sea exploration of minerals and hydrothermal vent fluid chemistry [85,86,88,89].

Plasma formation and target shielding. In both cases, plasma is formed at a linear absorbing solid target immersed in water (or another liquid). For discussing the ablation of linear absorbing materials, we will in the following refer to the radiant exposure threshold rather than to the irradiance threshold. The surrounding liquid features little linear absorption at the laser wavelength, and the optical breakdown threshold of the liquid, $F_{\mathrm{th,liquid}}$, lies above the ablation threshold of the target, $F_{\mathrm{th,target}}$. Thus, for pulse energies close to the ablation threshold of the target material, plasma formation is restricted to the target surface (Scenario I). However, at higher pulse energies, ionizing radiation from plasma formed at the target surface may ignite breakdown in the liquid, which then shields the target, as observed previously for breakdown in air ignited by plasma at a water/air surface [317] (Scenario II). Pure shielding would impair ablation in LAL and reduce the spectroscopic signal from the analyte in LIBS. However, the plasma formed in the liquid can, in conjunction with the target plasma also heat and melt the target and eject relatively large droplets [69,70,76]. Both scenarios are more strongly separated in LAL of metallic targets, where $F_{\mathrm{th,target}} << F_{\mathrm{th,liquid}}$ than in LAL of oxides, semiconductors and organic materials as well as in LIBS, where ablation thresholds may be closer to $F_{\mathrm{th,liquid}}$.

For process optimization, it is useful to know the thresholds for plasma-mediated target ablation and liquid breakdown as a function of pulse duration, wavelength and mode of laser operation. However, this knowledge alone is not sufficient to understand the conditions for ignition of liquid breakdown at the target surface and its propagation into the liquid during the laser pulse. Electrons are ejected from solid

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targets during plasma-mediated pulsed laser ablation [71,318-320] but the penetration depth of electrons into water is below 30 nm for energies up to 500 eV and the attenuation length of UV radiation remains below 20 nm for photon energies up to 27 eV (Fig. 14). For direct bremsstrahlung emission beyond 20 nm, the kinetic energy of the plasma electrons would have to exceed 27 eV, which would require electron temperatures > 300,000 K [321-324]. Such high electron temperatures can be reached under conditions, where a plasma skin layer forms [178,325] but this will not occur in pulsed laser ablation of metals for pulse energies below the liquid breakdown threshold. We conclude that for $F_L < F_{th,liquid}$ an ignition of liquid breakdown resembling the halo formation outside the laser cone angle in Fig. 7 (b)-(f) is not possible.

Inside the region traversed by the laser beam, another mechanism for breakdown ignition may come into play, which relies on radiation-induced generation of seed electrons for AI. The band structure of water features an intermediate state at the energy level of solvated electrons around 6.5 eV that lies below the lower edge of the conduction band at around 9.5 eV. Therefore, plasma initiation can proceed through excitation of water molecules above 6.5 eV with subsequent electron abstraction and upconversion into the CB [36,37]. Electron emission from the target plasma will first initiate liquid breakdown in a several nanometer thick layer next to the target. The interplay of IBA and impact ionization produce CB electrons in this layer with a kinetic energy spectrum reaching up to ≈ 14.5 eV (the energy required for impact ionization), with average kinetic energy around 6.8 eV [149]. Bremsstrahlung from these electrons with photon energies larger than 6.5 eV can excite water molecules and initiate plasma formation in adjacent liquid. Figure 14 shows that the optical penetration depth is as large as 10 cm for 6.5-eV photons and drops rapidly to 30 nm when the photon energy increases to 8.5 eV. Thus, the penetration depth suffices to drive a radiation-ionization wave propagating through creation of seed electrons for AI in upstream regions, from which then again UV radiation is emitted. Our hypothesis is corroborated by the relatively large extent of nanosecond plasmas at the BPL threshold in bulk liquid (Fig. 8) [108]. Once plasma is formed, the breakdown process becomes here independent of the creation of initial electrons by multiphoton ionization, and the threshold is lowered. Therefore, the plasma grows further than predicted by the moving breakdown model, where the plasma extent is defined by iso-intensity lines at I_{th}. Nanosecond pulse durations favor the formation of a radiation-ionization wave as they give time for an AI-driven buildup of ionization strong enough for significant bremsstrahlung emission providing seed electrons in new liquid regions. The possible depth of the liquid breakdown region depends on the AI rate, which grows with irradiance and wavelength (Section 2.1).

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For $F_{th,target} < F_L < F_{th,liquid}$, it remains to be explored under which conditions laser energy is merely absorbed by the target and an expanding target plasma (Scenario I) [75,76,78], and when plasma grows also by ignition of liquid breakdown through ionizing radiation from the primary plasma (Scenario II).

To date, most experimental studies on LAL and LIBS present plasma images only for times ≥ 100 ns and with limited spatial resolution [78,90,91,326-329]. An exception with regard to temporal resolution is the study by Song et al. on LAL of a metallic glass with image series taken at 100 Mio frames/s [330]. For a pulse energy well below the breakdown threshold of bulk water, the authors observe a thin luminescent plasma layer at the target surface and point-like plasmas in the liquid within the 3-mm laser beam that may be formed at nanoparticles produced earlier. In the context of LIBS, the early plasma phase and initial bubble expansion has been investigated in free liquid [88,331] but not yet at solid targets. With large pulse energies, the plasma volume seen on photographs at $t \geq 100$ ns is often very large but it is unclear whether this is due to an expansion of the target plasma or whether liquid breakdown is involved. High-resolution plasma photography in side view could elucidate details of plasma formation and the possible movement of a radiation-ionization wave into the liquid. Time-gated photography with a fast ICCD camera can further elucidate the temporal evolution – but without the spectral information available from CCD or CMOS color sensors.

Ablation rate and efficiency. Without plasma shielding, the ablation depth with ultrashort laser pulses up to ≈ 20 ps exhibits an approximately logarithmic dependence on pulse energy, $d_{abl} \propto \ln E_L$ [177,187,332]. The formation of a highly absorptive plasma skin layer in the target reduces the ablation rate at high radiant exposures [177,178]. Under conditions, for which the $d_{abl} \propto \ln E_L$ law holds, the ablation efficiency (ablated mass per unit energy) has a maximum at 2.7 times the ablation threshold [75,188]. This favorable condition is likely still part of the ablation regime, where liquid breakdown can be avoided. In nanosecond ablation, the energy dependence of the ablation rate is more complex, and ablation efficiency increases with the transition from surface vaporization to phase explosion regime, which goes along with substantial ionization of vaporized metal atoms [322]. The freed electrons will favor the ignition of nearby liquid breakdown.

Connection between ablation and bubble dynamics. The thermodynamic conditions during cavitation bubble initiation, expansion, and collapse influences the particle size distribution in LAL [74,75,81,90-94,333-336]. Valuable mechanistic insights into the initial ablation and bubble formation phase have been obtained through atomistic simulations for metallic targets [81,94,337]. Three distinct nanoparticle generation mechanisms were identified in the simulations: (1) the formation of a thin

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